TEST CAMPAIGNS: DP STEEL SHEET

One material (one steel sheet), per strain rate. Source: Beerli et al. (2026), Int. J. Impact Eng. 215, 105761, sections 2.2, 4.1 and 4.2. Lab names of the catalogue (10/11/2026). RD = rolling direction.

Not sure: inferred from the paper or to choose. The last column gives the question (end of the document).

1. SLOW RATE (QUASI-STATIC)

UT: 0.001 /s · other specimens: 6.6e-6 m/s · 100 kN hydraulic testing machine

SPECIMEN	ORIENTATIONS TO RD (°)	NO. ORIENT.	REPEATS	NUMBER	QUESTION
Tension-type specimens (waterjet)					
UT40	0, 15, 30, 45, 60, 75, 90	7	3	21	
NT20	0, 45, 90	3	3	9	
NT06	0, 45, 90	3	3	9	
CHD48 (CH5)	0, 45, 90	3	3	9	Q1, Q8
MNAK_RECT_FULL (punch)	0, 45, 90	3	3	9	Q7
MNAK_RECT_HOLE (mini-Nakazima)	0, 45, 90	3	3	9	Q7
Subtotal				**66**	
Shear (wire EDM)					
SHMD	0, 45, 90	3	2	6	Q6
Subtotal				**6**	
TOTAL				**72**	

2. MEDIUM RATE (INTERMEDIATE)

UT: 1 /s · other specimens: 6.6e-3 m/s · Instron 8801 (100 kN)

SPECIMEN	ORIENTATIONS TO RD (°)	NO. ORIENT.	REPEATS	NUMBER	QUESTION
Tension-type specimens (waterjet)					
UT40	0	1	3	3	Q3
NT20	0	1	3	3	Q3
NT06	0	1	3	3	Q3
CHD48 (CH5)	0	1	3	3	Q1, Q3, Q8
Subtotal				**12**	
Shear (wire EDM)					
SHMD	0	1	3	3	Q3, Q6
Subtotal				**3**	
TOTAL				**15**	

3. HIGH RATE

UT: 100 /s · other specimens: about 1.4 m/s · split Hopkinson tensile bar

SPECIMEN	ORIENTATIONS TO RD (°)	NO. ORIENT.	REPEATS	NUMBER	QUESTION
Tension-type specimens (waterjet)					
UT19 (small UT)	0	1	3	3	Q3, Q5
NT20	0	1	3	3	Q3
NT06	0	1	3	3	Q3
CHD48 (CH5)	0	1	3	3	Q1, Q3, Q8
Subtotal				**12**	
Shear (wire EDM)					
SHMD	0	1	3	3	Q3, Q6
Subtotal				**3**	
TOTAL				**15**	

All rates: 72 + 15 + 15 = **102 specimens per material**.

QUESTIONS FOR THE SUPERVISORS

Q1 **MATERIAL.** Which steel grade (and thickness) is this campaign for? This decides the central-hole size (CH5 or CH8) and the shear geometry.

Q2 **SHEETS.** Should the medium- and high-rate specimens (24 tension + 6 shear) come from the same 600 × 500 sheet, or from another sheet? The 600 × 500 sheet is already full with the slow-rate campaign (61 specimens).

Q3 **REPEATS AT MEDIUM AND HIGH RATE.** 3 per specimen? The paper does not say; 3 is inferred from its 150 tests (5 materials × 2 rates × 5 specimen types × 3).

Q4 **ORIENTATIONS AT MEDIUM AND HIGH RATE.** Only 0° to the rolling direction, as in the paper?

Q5 **HIGH-RATE UNIAXIAL TENSION.** Is UT19 the small UT specimen of the paper (gauge 15 × 5 mm)?

Q6 **SHEAR.** Only SHMD (the paper uses the "medium ductility" geometry), or all three LD / MD / HD? Orientations 0, 45, 90° (paper) or also 135° (2025 plan)? 2 repeats (paper) or 3? (See also the e-mail about pre-cut blanks.)

Q7 **PUNCH AND MINI-NAKAZIMA.** Is MNAK_RECT_FULL the punch specimen (equi-biaxial) and MNAK_RECT_HOLE the dihedral mini-Nakazima (plane strain)? 0, 45, 90° × 3 for both? (The current 600 × 500 order has MNAK_RECT_FULL at 0° only.)

Q8 **CENTRAL HOLE.** Cut at 4.8 mm and then reamed to 5 mm? Is an 8 mm version needed (CH8, for DP1180 / DP1470)?

Q9 **CLAMPING ENDS.** The paper uses a different clamping end for each rate. Are the same fir-tree ends used at all rates?

Q1 and Q2 first: they decide the central-hole type, the shear geometry and the number of sheets to cut.